PRISM: Personalized Prediction of Long Screen-Off Intervals on Smartphones

Yichen Peng[1], Shuaiming Zhu[2], Changhua Pei[2], Jingjing Li[2], Xiao Liang[1]

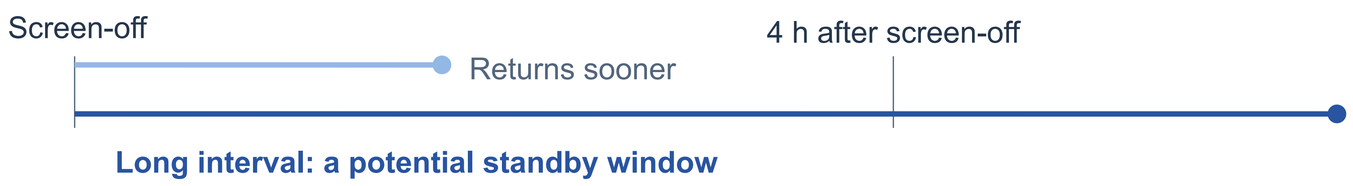
[1] The High School Affiliated to Renmin University of China
[2] Computer Network Information Center, Chinese Academy of Sciences

1 Problem Setup

Deep standby could **reduce background energy use**.
Predicting long intervals helps avoid premature entry and recovery overhead.

Research Question

Will the next screen-on occur **more than four hours** after screen-off?



Dataset

~4.9M screen-off events from **53,122** smartphone users.
31.96% last more than 4h.

Prediction protocols

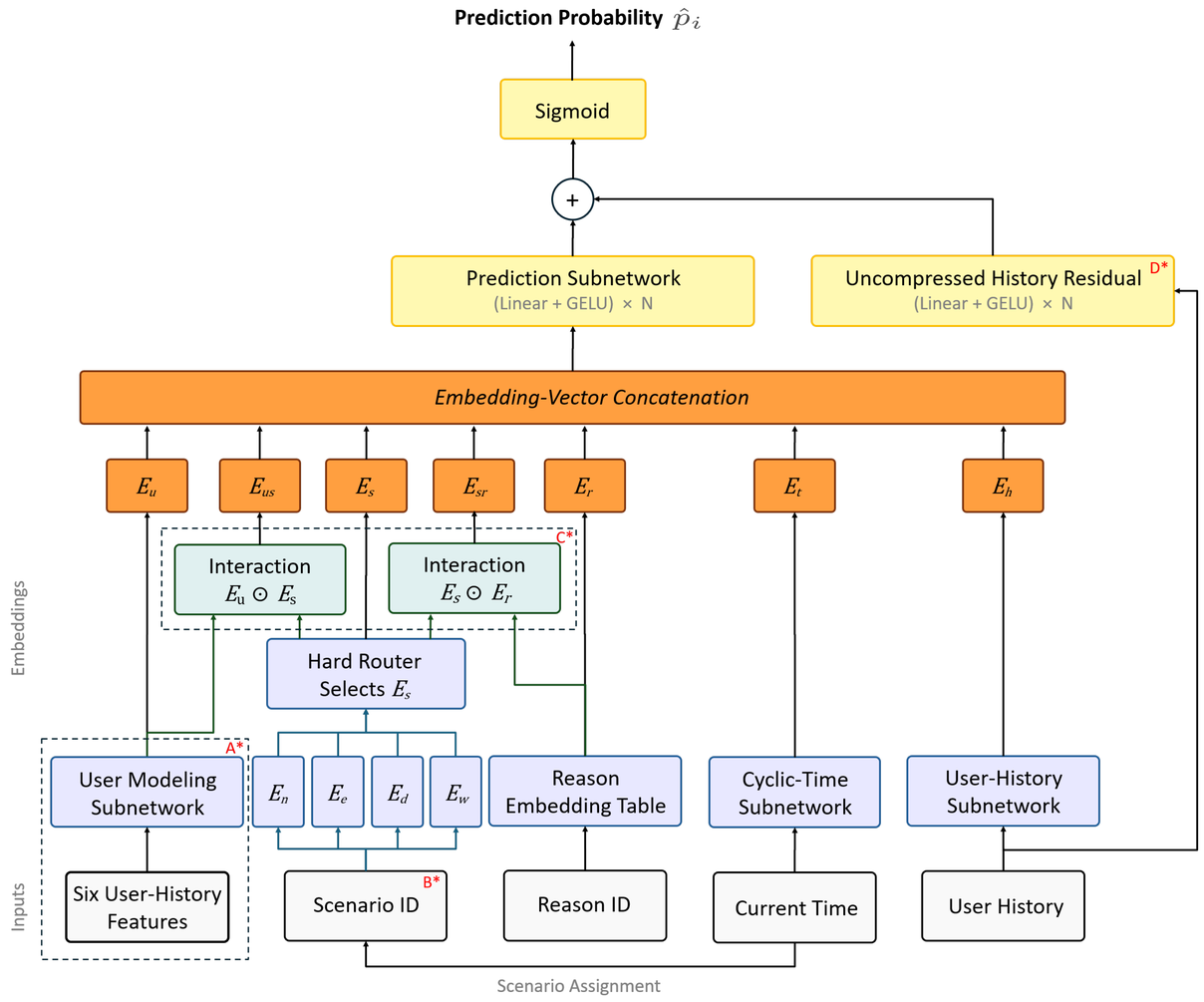
Immediate: at screen-off.
Delayed: after 0.5 or 2.5h, depending on scenario.

2 The PRISM Model

Personalized Reason-Individual-Scenario Model

A **fast, compact, accurate predictor** of long screen-off intervals for potential deep standby.

Highest accuracy and F1 among evaluated models under both protocols.



A Continuous user profiles

Six historical statistics summarize habits and recent behavior. A jointly trained neural network maps them to a continuous user profile that personalizes the prediction.

Ablation results: highest F1 among five profile representations under both protocols.

B Scenario ID: Four daily scenarios

Daily routines suggest four scenarios: Night, Weekday Day, Weekday Evening and Weekend. We set Night to 23:00-07:00 by maximizing the long-off rate over 8-hour windows. We split weekday day and evening at 17:00 by maximizing Bernoulli log-likelihood.

Ablation results: highest immediate F1 among four scenario representations.

C Two explicit interactions

We inspect positive-rate tables for three pairs. Profile-scenario and scenario-reason rates vary across both factors. Profile-reason rates mainly depend on reason, so we model the first two pairs.

Ablation results: highest F1 among eight interaction subsets under both protocols.

D Uncompressed history residual

Three histories summarize completed user events: overall, with the same reason, and in the same scenario. Their full statistics feed a residual branch to retain details that compression may lose.

Ablation results: +0.266 / +0.187 pp F1 vs. no residual, immediate / delayed.

Each ablation varies one design choice while keeping all other settings fixed. See Tables 2–5 in the paper for full results.

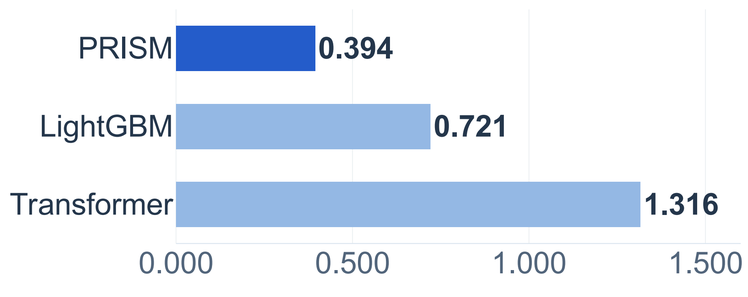
3 Results: Best Accuracy and F1, Fast Inference, and Gains Across All Scenarios

Chronological split: 70% training / 10% validation / 20% test.

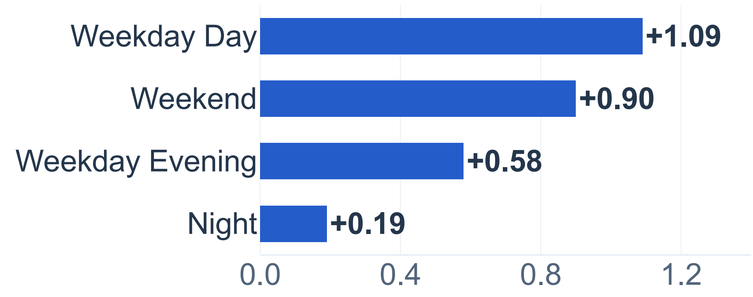
Positive-class F1 (%)

Model	Immediate	Delayed
Bayes	80.31	85.92
LightGBM	82.19	89.27
Transformer	82.25	88.98
PRISM	**82.96**	**89.50**

Inference time per event (ms)



F1 gain vs best baseline (pp, immediate)



88.71% / 92.89%
Immediate / Delayed accuracy

1.8× / 3.3× faster than
LightGBM / Transformer

4 / 4 scenarios improved
vs. LightGBM / Transformer

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